DIGITAL FORENSICS

FINAL INVESTIGATION REPORT

Hidden Metadata & Steghide Extraction

Case Reference No.	*DFIR-2026-002*
Investigator	*Julian Derry*
Date of Analysis	*February 17, 2026*
Prepared For	*Public*
Classification	*Public*

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1. Overview / Case Summary

This challenge involved analyzing an image file (img.jpg) to recover a hidden payload and extract the final flag. The investigation focused on metadata analysis, Base64 decoding, and steganographic extraction.

Target File: *img.jpg*

Objective: *Recover the hidden secret and the final flag*

Case Information

Investigator Name	*Julian Derry*
Date of Analysis	*February 17, 2026*
Case Reference Number	*DFIR-2026-001*
Client / Requesting Party	*HIVE CONSULT*
Platform(s) Examined	*KALI LINUX*
Tools Used	*Kali Linux, ExifTool, Steghide, base64(GNU Core Utilities)*
Report Classification	*Public*

2. Metadata Examination

The first step was to inspect the image metadata using ExifTool.

Command: *exiftool img.jpg*

```
┌──(kali㉿kali)-[~/Desktop]
└─$ exiftool img.jpg
ExifTool Version Number         : 12.76
File Name                       : img.jpg
Directory                       : .
File Size                       : 73 kB
File Modification Date/Time     : 2026:02:17 09:48:34-06:00
File Access Date/Time           : 2026:02:17 09:48:54-06:00
File Inode Change Date/Time     : 2026:02:17 09:48:54-06:00
File Permissions                : -rw-------
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 1
Y Resolution                    : 1
Comment                         : c3RlZ2hpZGU6Y0VGNmVuZHZjbVE9
Image Width                     : 640
Image Height                    : 640
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 640x640
Megapixels                      : 0.410
```

The output revealed a suspicious Comment field:

Result: *c3Rle2hpZGU6cEF6endvcmV9*

This string appeared to be Base64-encoded.

3. Base64 Decoding (Layer 1)

Decode the metadata value:

Command: *echo c3Rle2hpZGU6cEF6endvcmV9 | base64 -d*

```
┌──(kali㉿kali)-[~/Desktop]
└─$ echo "c3RlZ2hpZGU6Y0VGNmVuZHZjbVE9" | base64 --decode
steghide:cEF6endvcmQ=
```

Result: *steghide:cEF6endvcmV=*

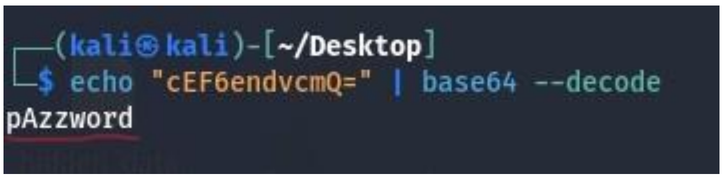
Interpretation

- steghide → indicates the steganography tool likely used.
- cEF6endvcmV= → another Base64-encoded value.

4. Base64 Decoding (Layer 2)

Decode the metadata value:

Command: *echo cEF6endvcmV= | base64 -d*



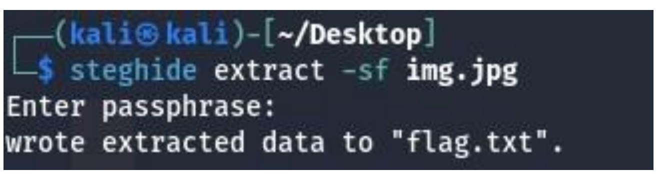
Result: *pAzzword*

This is the **Steghide passphrase**

5. Extract the hidden Payload

Use the recovered passphrase with Steghide:

Command: *steghide extract -sf img.jpg*



When prompted for the passphrase, enter:

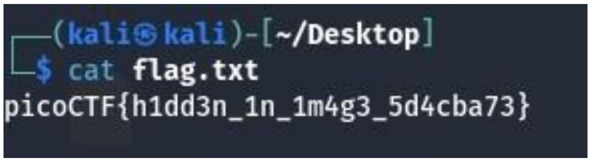
pAzzword

A hidden text file was extracted successfully.

4. Recover the Flag

Read the contents of the extracted file:

Command: *cat secret.txt*



Results: *picoCTF{h1dd3n_1n_1m4g3_5d4cba73}*

Final Flag

picoCTF{h1dd3n_1n_1m4g3_5d4cba73}

Investigation Summary

Stage	Action	Result
Metadata analysis	exiftool img.jpg	Found encoded comment
Base64 decode #1	Decoded metadata value	steghide:cEF6endvcmV=
Base64 decode #2	Decoded passphrase value	pAzzword
Steghide extraction	Extract hidden file	Success
Flag recovery	Read extracted file	picoCTF{h1dd3n_1n_1m4g3_5d4cba73}

8. Analyst Notes

This challenge demonstrates how valuable **image metadata** can be during forensic and steganographic investigations. A seemingly harmless metadata field contained the clues required to recover both the extraction passphrase and the final hidden flag.